The Tholonic Model as a Candidate Alternative: to the Standard Model: Structural Derivations and Parameter Reduction

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Abstract

The Standard Model of particle physics is the most precisely tested physical theory in history. It is also explicitly incomplete: it requires 17 independently postulated quantum fields and 19 free parameters inserted from measurement rather than derived from principle. A candidate replacement theory must satisfy a specific criterion: it must reduce the axiom count and derive at least some of the free parameters from structural geometry. This paper evaluates the tholonic model against that criterion. The tholonic framework proposes a single generative structure, the N-D-C triad (Negotiation, Definition, Contribution), as the irreducible unit of all stable, self-sustaining organization. Eight prior papers in this series establish the mathematical grounding of this framework, prove the structural necessity of exactly three roles (Paper 3), derive five classical mathematical constants from the triadic recursion (Paper 1), and show that the atom's measurable properties, including quark occupancy structure, color charge, the virial theorem, tetrahedral orbital geometry, and the Koide formula's unexplained 2/3 lepton mass ratio, are structural predictions of the tholonic framework rather than independent empirical accidents (Paper 8). This paper synthesizes those results into a direct comparison with the Standard Model. The tholonic framework addresses five of the Standard Model's unexplained structural facts (three fermion generations, three color charges, three spatial dimensions, quark confinement, and the Koide 2/3) through structural arguments that require no new free parameters. A systematic search over tholonic-constant expressions identifies $4 \cdot e^5 \cdot (\ln 2)^4 = 137.035865$ as the leading numerical candidate for $1/\alpha$, with an error of -0.98 ppm and all exponents being fixed thologram structural values, not fitted parameters. A second open derivation targets the proton/electron mass ratio via the known approximation $6\pi^5 \approx 1836.12$ and its connection to the tholonic N-weight. The paper includes a systematic parameter comparison table, explicit falsifiability conditions, and a prioritized roadmap for the derivations that would, if completed, constitute the strongest available confirmation of the framework. The tholonic model is speculative; all claims should be understood as structurally motivated hypotheses requiring independent verification.

1 Introduction

1.1 The Standard Model's achievements

The Standard Model of particle physics is among the most successful scientific theories ever constructed. It correctly predicts the anomalous magnetic moment of the electron to eleven significant figures, the existence and mass of the W and Z bosons, the top quark, and the Higgs boson before any of them were directly measured. Its quantum field theoretic framework has been validated across energies ranging from atomic-scale phenomena (~ 1 eV) to the highest-energy collider experiments ($\sim 10^4$ GeV). No confirmed experimental deviation from Standard Model predictions has been established despite decades of intensive search.

This paper does not dispute any of those achievements. The tholonic model does not predict that any currently confirmed Standard Model result is wrong. Its claim is structural and prior: that the Standard Model, as presently constituted, is not a complete explanation of the physical world, and that the tholonic framework addresses some of its structural gaps in a way that reduces rather than multiplies explanatory posits.

1.2 The Standard Model's structural limitations

The Standard Model's incompleteness is not a matter of experimental dispute. It is openly acknowledged by its practitioners and constitutes one of the primary motivations for theoretical work in fundamental physics. The specific deficits relevant to this paper are:

The parameter problem. The Standard Model requires 19 free parameters (26 including the full neutrino sector). None of these is derived from the theory's structure. Each is measured and inserted. A table of the 19 parameters is given in Section 5. The electromagnetic coupling constant $\alpha \approx 1/137.036$, the ratio of the proton mass to the electron mass (≈ 1836.15), and the Koide formula's ratio $Q = 2/3$ for lepton masses are among the most prominent examples. Richard Feynman described α as "one of the greatest damn mysteries of physics." No Standard Model calculation produces its value; it is taken from experiment.

The field proliferation problem. The Standard Model requires 17 independently postulated quantum fields: one for each of the 6 quarks, 3 charged leptons, 3 neutrinos, the photon, the W^+, W^-, Z^0, gluon, and Higgs boson. These are not derived from each other. Each is a separate axiom. A theory that replaced the 17 axioms with one and derived the particle spectrum from the geometry of that one field would, by any standard measure of explanatory economy, be a better theory.

The structural silence problem. The Standard Model has no answer to three of the most basic structural questions about the physical world: Why are there exactly three fermion generations? Why are there exactly three color charges? Why are there exactly three spatial dimensions? These are observed facts that the Standard Model accommodates without explaining.

The confinement problem. The Standard Model derives quark confinement from the running of the strong coupling constant in QCD. This is correct and well-tested. But it does not explain why the confinement principle generalizes: any partial combination of less than three quarks is either transient or confined. The QCD explanation is force-specific; the generalization across scales requires additional argument.

1.3 The criterion for a better theory

Following the standard of explanatory economy articulated by Einstein in his pursuit of a unified field theory, a candidate replacement or successor to the Standard Model must satisfy at least one of:

1. Reduce the number of independently postulated fields (axiom reduction).
2. Derive at least one currently free parameter from structural geometry (quantitative derivation).
3. Explain at least one currently unexplained structural fact (structural prediction).

This paper evaluates the tholonic model against all three criteria. The claim is not that the tholonic model currently satisfies all three comprehensively. The claim is that it satisfies criterion 3 on five counts, is within 1 ppm on criterion 2 for the fine structure constant, and provides a principled path toward criterion 1.

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2 The Tholonic Framework: Summary of Prior Results

2.1 The core claim

The tholonic model's foundational claim is that all stable, self-sustaining structures in nature are instances of a single triadic pattern called the N-D-C triad: Negotiation (the emergent equilibrium), Definition (the bounding/constraining role), and Contribution (the integrating/accumulating role). This is a structural claim, not a metaphysical one. It asserts that any entity that persists in time does so because it simultaneously satisfies three functional roles: something that limits its form (D), something that flows and integrates across that form (C), and a stable balance state that emerges when D and C are in equilibrium (N).

Paper 3 of this series proves formally that three roles are the minimum for non-trivial convergent recursion. A system with fewer than three roles is either trivially stable (constant, no dynamics) or divergent (no stable equilibrium). The proof is constructive and does not depend on any physical assumption.

2.2 N-D-C role assignments and the thologram

The thologram is the geometric realization of the N-D-C triad at successive levels of recursion. It is a triangular structure with six labeled points: three outer vertices carrying the N, D, C roles and three inner midpoints carrying complementary roles. Three distinct numerical patterns arise from this structure:

Binary vertex values (1, 2, 4): The outer vertices carry $N = 2^0 = 1$, $D = 2^1 = 2$, $C = 2^2 = 4$. These reflect the roles' positions in a binary state space.

Axis multipliers (2, 3, 5): Summing vertex values along each directed axis yields $14 = 7 \times 2$ (N axis), $21 = 7 \times 3$ (C axis), $35 = 7 \times 5$ (D axis). The axis multipliers are the tholographic seeds for the $\pi/4$ branch derived in Paper 1.

Role weights (2-3-6): The structural role weights are D = 2, C = 3, N = 6, with D:C weight ratio = 2/3.

These three patterns are not mutually exclusive. A given physical correspondence may be illuminated differently by each, and the appropriate pattern for a given context is determined by which aspect of the tholonic structure is being examined.

2.3 Results from the paper series

The eight prior papers establish the following results on which this paper builds:

- **Paper 1:** Five classical constants ($\pi/4$, ϕ, e, $\sqrt{2}$, $\ln 2$) emerge as limits of the tholonic recursive system from its own internal geometry, without separate derivations for each.
- **Paper 3:** Three functional roles are the proved minimum for non-trivial convergent recursion. Any fewer roles produces either a constant system or a divergent one.
- **Paper 6:** The integers 1, 2, and 3 have qualitatively distinct structural roles in minimal recursive systems, providing a formal basis for the N-D-C assignment.
- **Paper 7:** The N-D-C framework has been independently rediscovered by Cambridge Semantics through two decades of production engineering of large-scale knowledge graphs, providing an empirical convergence that is not constructed from tholonic premises.
- **Paper 8:** The atom is measurably a tholon. Six specific correspondences are established, graded by strength, and compared with the Standard Model's explanatory coverage.

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3 Structural Axiom Reduction

3.1 The three-role minimality theorem

Paper 3's central result is that the N-D-C triad is not an arbitrary choice of three roles but the unique minimum structure for convergent recursive self-organization. The theorem states: any recursive system that achieves non-trivial stable convergence requires at least three distinct functional roles in the N-D-C pattern. Systems with fewer roles either do not converge (no D to bound C) or converge trivially to a constant (no C to integrate). Systems with more than three roles are either decomposable into N-D-C triads or unstable due to over-constraint.

This result has direct consequences for how many of the Standard Model's separately postulated structures are actually independent. If the three fermion generations, three quark colors, and three-quark composition of hadrons are all consequences of the same three-role minimality requirement applied at different scales, then they are not three independent axioms. They are one axiom, instantiated three times.

3.2 Three color charges as a tholonic structural prediction

Quantum chromodynamics requires exactly three color charges (red, green, blue) as a SU(3) gauge symmetry. The Standard Model postulates SU(3) as the gauge group of the strong force. It does not derive why the gauge group is SU(3) rather than SU(2) or SU(4).

The tholonic argument for three colors is structural: a stable quark configuration requires three N-D-C roles, one at each vertex of the tholographic trigram. Each vertex carries one color. Color neutrality, the observed requirement that stable hadrons carry zero net color charge, is the tholonic requirement that every stable N-D-C configuration must be balanced across all three roles. An unbalanced configuration (one or two colors dominant) is the tholonic equivalent of an unstable partial tholon.

The tholonic model does not replace the QCD derivation of SU(3); it provides a prior structural reason for why a three-generator gauge symmetry is the correct framework. The convergence of SU(3) from QCD and three-role minimality from the tholonic model is itself a form of consilience.

3.3 Three fermion generations as a structural prediction

The Standard Model accommodates three generations of fermions (electron family, muon family, tau family; up/down family, charm/strange family, top/bottom family) but provides no explanation for why there are exactly three. No fourth-generation charged fermion has been found; precision electroweak measurements at LEP constrain the number of light neutrino species to $N_\nu = 2.9840 \pm 0.0082$, consistent with exactly three.

The tholonic prediction follows directly from Paper 3's minimality theorem: if the three-role requirement applies at the generation level of the fermion spectrum (each generation playing one of the N, D, C roles in the fermion tholon), then exactly three generations is a structural necessity, not a numerical coincidence. The N generation (first: electron, up/down) is the lightest and most stable; the D generation (second: muon, charm/strange) is the bounding/constraining generation; the C generation (third: tau, top/bottom) is the heaviest, most unstable, most integrating generation.

The asymmetry in stability, with lighter generations more stable and heavier ones unstable on cosmological timescales, is consistent with the tholonic stability gradient: the N role (first generation) is the most stable equilibrium state, D (second) is stable but transient under weak decay, and C (third) is the least stable, most contributing role, characteristically unstable and decaying to the first generation under normal conditions.

A fourth generation is not prohibited by experiment at high mass, but is constrained by the LEP measurement cited above for light neutrinos. The tholonic model's structural prediction is that no fourth generation exists at any mass, because the three-role structure is complete. This is falsifiable by discovery of a fourth-generation fermion.

3.4 Quark confinement as incomplete-tholon instability

QCD derives quark confinement through the running of the strong coupling constant α_s, which increases with distance (infrared slavery). As two quarks are pulled apart, the energy in the gluon field between them grows linearly with distance until it exceeds the energy required to create a new quark-antiquark pair from the vacuum. Confinement is the result: free quarks are never observed.

The tholonic structural argument provides an independent reason for the same fact. A single quark occupies one N-D-C role in the tholographic trigram. It is, by definition, an incomplete tholon: it has one role but lacks the two remaining roles required for a stable N-D-C configuration. An incomplete tholon is structurally unstable. It cannot persist as an isolated entity; it must either complete its structure (combine with other quarks to form a hadron) or not exist as a stable isolated object.

The tholonic and QCD explanations are complementary rather than competing. The QCD explanation specifies the mechanism (running coupling, string formation, pair creation). The tholonic explanation identifies the structural reason that generalizes beyond the specific force: any partial tholon is unstable, regardless of the specific interaction governing it. This generalization predicts that confinement-like behavior should appear in any system where the underlying structure is tholonic and partial configurations are accessible. The quark system is the clearest known instance.

3.5 Three spatial dimensions as a tholonic constraint

The Standard Model is formulated in 3+1 spacetime but provides no explanation for why physical space has exactly three dimensions. The tholonic model offers a structural argument: the tetrahedron is the minimum structure that encloses a volume and the minimum three-dimensional closure of the N-D-C trigram. In fewer than three spatial dimensions, no volume can be enclosed and the tholographic recursion cannot achieve three-dimensional self-similarity. In more than three spatial dimensions, the tetrahedron is no longer the minimal enclosing structure and the tholonic three-role closure is not uniquely defined.

If the tholonic framework formally proves that stable recursive N-D-C structure requires exactly three spatial dimensions for volumetric closure, this constitutes a structural derivation of a fact the Standard Model cannot address. This argument is currently informal; a formal proof is identified as an open derivation in Section 7.

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4 Quantitative Parameter Derivations

4.1 The fine structure constant: leading candidate

The fine structure constant α governs the strength of the electromagnetic interaction. Its measured value (CODATA 2022) is:

$$\frac{1}{\alpha} = 137.035999206(11)$$

This value appears as a free parameter in the Standard Model. It is measured and inserted. No Standard Model calculation derives it from the theory's structure. Arthur Eddington famously argued on numerological grounds that $1/\alpha$ is exactly 137; Feynman described α as "one of the greatest damn mysteries of physics." Prior numerological attempts are cautionary context: Eddington's derivation failed because he chose a structure to fit the number rather than deriving the number from the structure.

The tholonic approach imposes a structural constraint before any comparison with measurement: the expression must be built exclusively from the five tholonic constants ($\pi/4$, ϕ, e, $\sqrt{2}$, $\ln 2$) and thologram structural integers (the axis multipliers 2, 3, 5; the outer vertex values 1, 2, 4; and the role weights 2, 3, 6), with all exponents being those fixed integers. No fitting is permitted after the fact.

A systematic computational search over all such expressions identifies a leading candidate:

$$\frac{1}{\alpha} \approx 4 \cdot e^5 \cdot (\ln 2)^4 = 137.035865 \quad \text{(error } -0.98 \text{ ppm)}$$

Since $(\sqrt{2})^4 = 2^2 = 4$ exactly, this is equivalently:

$$\frac{1}{\alpha} \approx e^5 \cdot (\sqrt{2})^4 \cdot (\ln 2)^4$$

a product of three tholonic constants with integer exponents. Its tholonic structure is:

$$\frac{1}{\alpha} \approx C_{\text{outer}} \cdot e^{D_{\text{axis}}} \cdot (\ln 2)^{C_{\text{outer}}}$$

where $C_{\text{outer}} = 4 = 2^2$ is the C outer vertex value, $D_{\text{axis}} = 5$ is the Definition axis multiplier, and the same $C_{\text{outer}} = 4$ reappears as the exponent of $\ln 2$. All three structural numbers are fixed thologram values. No alternating series with tholonic seeds, no two-constant product with integer exponents up to ± 8, and no four-constant product within the searched range came within 200 ppm of $1/\alpha$. This candidate is the unique expression in the three-tholonic-constant space within 1 ppm of the measured value.

The structural reading of the formula encodes the thologram's D-C relationship: the D axis multiplier (5) governs the exponential rate via e, while the C outer vertex value (4) appears twice, as both the scale factor and the power of $\ln 2$. The C role is the accumulating, integrating component; $\ln 2$ is the tholonic constant associated with logarithmic (integrating) convergence (Paper 1, Class B branch). The D role is the bounding component; e raised to the D-axis power sets the overall magnitude. Whether this structural reading can be derived from role axioms rather than observed numerically is the central open theoretical question.

Status of the claim. The leading candidate is 0.98 ppm from the measured value. It is not exact at CODATA precision (0.15 ppb). It is the unique candidate in the constrained search space. A tholonic derivation of α should either produce an expression in the neighborhood of $C_{\text{outer}} \cdot e^{D_{\text{axis}}} \cdot (\ln 2)^{C_{\text{outer}}}$ or explain why that expression coincides to 1 ppm with the measured value while being structurally motivated.

Secondary candidate. A secondary expression $4\pi^3 + \pi^2 + \pi = 137.036304$ (error $+2.22$ ppm) can be read as $C \cdot \pi^{D+N} + N \cdot \pi^D + N \cdot \pi^N$ with outer vertex values $N = 1$, $D = 2$, $C = 4$. It is 2.3 times less accurate than the leading candidate and the D vertex is absent from the prefactors, which requires structural justification. It is listed here for completeness.

4.2 The proton/electron mass ratio

The ratio $m_p/m_e \approx 1836.15267$ is not derived in the Standard Model. The empirical approximation $6\pi^5 \approx 1836.118$ is known but unexplained. In the tholonic assignment for the hydrogen atom (Paper 8, Section 4), the proton occupies the D role, the electron occupies the C role, and the stable orbital (N state) emerges from their balance. The role weights in the 2-3-6 pattern assign N = 6, D = 2, C = 3.

The approximation $6\pi^5 \approx 1836$ has a natural tholonic reading: the N weight (6) multiplied by π (the tholonic constant most closely associated with the N recursion limit, the $\pi/4$ branch from Paper 1) raised to the D-axis power (5). Written with thologram structural integers:

$$\frac{m_p}{m_e} \approx N_{\text{weight}} \cdot \pi^{D_{\text{axis}}} = 6 \cdot \pi^5 = 1836.118 \quad \text{(error } -0.019\%\text{)}$$

This is a 190 ppm approximation, worse than the fine structure constant candidate but structurally analogous: both use the D-axis multiplier (5) as an exponent and thologram role integers as prefactors. The coincidence is suggestive. A derivation from role axioms would need to show why the D-axis integer appears as the exponent of π in the ratio of the D-role mass to the C-role mass. This is an open derivation.

4.3 The Koide formula: D:C weight ratio and lepton masses

The Koide formula (discovered empirically in 1982) states that the three charged lepton masses satisfy:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$$

This equality holds to current measurement precision ($Q_{\text{lepton}} = 0.66659$ using PDG 2024 values). The Standard Model has no derivation of this value; it is one of the most prominent unexplained numerical patterns in particle physics.

The tholonic 2-3-6 role weights assign D = 2 and C = 3, giving a D:C weight ratio of exactly 2/3. If the three lepton generations occupy the N, D, C roles in the lepton tholon, the Koide formula's $Q = 2/3$ is numerically identical to the tholonic D:C weight ratio. The value is not imported from measurement; it is the structural weight ratio of the tholonic roles.

The quark triplet constraint. Paper 8 (Section 8.1) reports the Koide Q values for the quark triplets using PDG 2024 masses:

$$Q_{uct} \approx 0.849 \quad Q_{dsb} \approx 0.731$$

Both values are substantially above 2/3. This is a partial falsification of the universal Koide claim. The tholonic identification $Q = D/C$ currently requires restriction to the lepton sector, with a structural justification for why quarks do not satisfy the same relation. Three candidate explanations are: (1) the Koide relation is a property of the lepton sector specifically, where color charge is absent; (2) quark mass running scheme dependence accounts for the deviation at a unified renormalization scale; (3) the structural derivation of the Koide form, once produced, will identify which triplets are expected to satisfy $Q = 2/3$. Until a structural derivation of the Koide quadratic mean form exists, this remains a suggestive correspondence rather than a full structural prediction.

4.4 The Tsirelson bound: summary of the Paper 8 derivation

Paper 8 (Section 3.4) derives the Tsirelson bound $2\sqrt{2}$ from tholonic first principles at two independent levels. The derivation is summarized here because it constitutes a completed structural derivation of a previously unexplained quantum bound.

From the real-virtual tholon pair. Paper 1 proves that the tholonic recursive system converges to $\sqrt{2}$ as one of its five classical limits. A complete tholon always comprises a real N-D-C configuration and its virtual complement. Both are full recursive triadic systems and both independently converge to $\sqrt{2}$. Their composite value is $\sqrt{2} + \sqrt{2} = 2\sqrt{2}$. An entangled particle pair is, in tholonic terms, a complete real-virtual tholon pair; the maximum achievable CHSH correlation is bounded by this composite convergence.

From pure angle geometry. The tholonic D-C balance angle is $\pi/4$ (the optimal angle where $\sin\theta = \cos\theta$), which is also the π-branch seed angle from Paper 1. At this angle, the maximum CHSH value is:

$$\mathrm{CHSH}_{\mathrm{max}} = 4\cos(\pi/4) = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

The CHSH sign pattern $(+1, +1, +1, -1)$ is derived from three tholonic axioms, the last of which (the D-C balance sign rule) is proved from Paper 3, Lemma 4.2, without reference to measurement scenarios or quantum mechanics. This is the first structural derivation of the CHSH operator form and Tsirelson value from non-quantum-mechanical first principles.

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5 Parameter Comparison: Standard Model vs Tholonic Model

5.1 Comparison table

The table below lists the Standard Model's major free parameters and structural posits alongside the tholonic model's current coverage. "Derived" means the tholonic model generates the value from structural geometry without a free parameter. "Structural" means the tholonic model provides a principled reason for the observed fact without importing it from measurement. "Open" means the tholonic model has a candidate or argument but lacks a completed derivation.

SM Parameter or Structural Fact	SM Status	Tholonic Status	Paper
Three fermion generations	Observed; no SM explanation	Structural: three roles are the minimum for stable recursion	3, 8

SM Parameter or Structural Fact	SM Status	Tholonic Status	Paper
Three color charges	Postulated (SU(3) axiom)	Structural: three N-D-C roles require three-fold balance	8
Three spatial dimensions	Observed; no SM explanation	Structural: tetrahedron is the minimum volumetric closure	8
Quark confinement	Derived from QCD running coupling	Structural: incomplete tholon is unstable	8
Koide $Q = 2/3$ (lepton masses)	Measured; no SM derivation	Derived from D:C weight ratio (lepton sector)	8
CHSH Tsirelson bound $2\sqrt{2}$	Derived from Hilbert space algebra	Structural derivation from tholonic first principles	8
Fine structure constant $1/\alpha \approx 137.036$	Measured; no SM derivation	Open: leading candidate $4 \cdot e^5 \cdot (\ln 2)^4$, error -0.98 ppm	this paper
Proton/electron mass ratio ≈ 1836	Measured; no SM derivation	Open: candidate $6\pi^5 \approx 1836.12$, error -190 ppm	this paper
Number of SM quantum fields (17)	Postulated independently	Open: reduction if color, generation, composition structure all follow from one tholonic recursion	3
Quark and lepton masses (12 parameters)	Measured; no SM derivation	Koide lepton sector derived; quark sector constrained by Q_{uct}, Q_{dsb} values	8
CKM mixing angles (4 parameters)	Measured; no SM derivation	Not yet addressed	–
Gauge coupling constants (3 parameters)	Measured; no SM derivation	α candidate above; weak and strong couplings not yet addressed	this paper
Higgs mass and quartic coupling (2 parameters)	Measured; no SM derivation	Not yet addressed	–

SM Parameter or Structural Fact	SM Status	Tholonic Status	Paper
Strong CP phase (near zero; unexplained)	Measured near zero; no SM derivation	Not yet addressed	–

5.2 What remains to be derived

The tholonic model's current position is strongest on structural facts (three generations, three colors, three dimensions, confinement, Koide 2/3, Tsirelson bound) and weakest on quantitative parameters where the Standard Model's fine-grained machinery (renormalization group, gauge structure, Higgs mechanism) is most powerful. The CKM mixing angles, Higgs parameters, and strong CP phase have no current tholonic address. The weak and strong coupling constants beyond α are not yet addressed quantitatively.

This is an honest accounting. A framework that addresses five of the Standard Model's structural silences and has a 1 ppm candidate for the most prominent free parameter is in a different position than a framework that merely claims consistency. But it is also not a complete alternative. It is a candidate alternative, with a specified list of open derivations that would, if completed, move it into the position of a genuine competitor.

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6 Falsifiability

The tholonic model is falsifiable. The following conditions are organized by type.

6.1 Strong falsification conditions

Fine structure constant. If a derivation from thologram geometry produces a value that differs from the measured 1/137.035999206, that specific derivation is falsified. If an exhaustive search over all tholonic-constant expressions with structural integers as exponents finds no candidate within, say, 10 ppm, the claim that α is a tholonic balance ratio is falsified at that scope.

Fourth fermion generation. If a fourth-generation charged fermion is discovered with mass accessible to current or near-future colliders, the structural prediction of exactly three generations is falsified.

Isolated free quark. If a stable, isolated free quark or color-nonsinglet hadron is confirmed experimentally, the incomplete-tholon instability argument for confinement is falsified.

Koide formula in lepton sector. If new precision measurements of the charged lepton masses show Q_{lepton} deviating from 2/3 by more than measurement uncertainty, the Koide correspondence is falsified at the level of precision measurement.

6.2 Structural constraints

N-D-C non-assignability. If a stable, self-sustaining, recursive physical system is found that cannot be consistently assigned an N-D-C structure at any scale, the universality claim of the tholonic framework requires explicit revision. The framework would need to specify which systems it applies to and why.

Koide quark sector. The quark triplet Q values ($Q_{uct} \approx 0.849$, $Q_{dsb} \approx 0.731$) already constitute a partial constraint. The tholonic model must either (a) restrict the Koide D:C correspondence to leptons with structural justification, or (b) demonstrate that a renormalization-group or color-corrected quark mass evaluation recovers $Q \approx 2/3$. Failure of both options weakens the Koide claim structurally.

6.3 What would constitute confirmation

A completed derivation of $1/\alpha$ from thologram geometry, producing a value within the CODATA measurement precision of 0.15 ppb, would constitute the single strongest available confirmation. It would demonstrate that a dimensionless physical constant is determined by the geometry of the tholonic recursion structure, which is precisely the criterion Einstein identified for a genuinely unified field theory.

A completed derivation of the proton/electron mass ratio from N-D-C role weights would constitute a second confirmation of the same type.

Discovery that the quark sector Koide Q values approach 2/3 at a unified renormalization scale would substantially extend the lepton-sector Koide correspondence to the complete fermion spectrum.

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7 The Path Forward

7.1 The fine structure constant derivation as the priority target

The fine structure constant derivation is the highest priority because it is the most testable and most consequential open item. The leading candidate $4 \cdot e^5 \cdot (\ln 2)^4$ is at 0.98 ppm. The gap between a numerical coincidence and a structural derivation is the gap between observing that the expression is close and proving that the structure of the thologram constrains α to take that value.

The direction for a structural derivation is suggested by the formula's own structure: $D_{\text{axis}} = 5$ appears as the exponent of e (the exponential growth constant), while $C_{\text{outer}} = 4$ appears both as the prefactor and as the exponent of $\ln 2$ (the logarithmic integration constant). The D-C duality in the exponent structure mirrors the D-C functional duality of the tholonic roles. A derivation from role axioms would need to show why the electromagnetic coupling strength, which governs how strongly charged fields interact, is determined by the ratio of the thologram's D-axis exponential scaling to its C-role logarithmic convergence.

One approach is dimensional: the fine structure constant is dimensionless, a ratio. Dimensionless quantities in physics characterize the balance between competing tendencies.

The tholonic model is precisely a theory of how competing tendencies (D and C) reach a stable balance (N). An electromagnetic coupling of $\alpha \approx 1/137$ says that the electromagnetic force is very weak compared to the fundamental unit of quantum charge. In tholonic terms, this is a statement about how far from the D-C balance point the electromagnetic interaction sits in the thologram. The derivation challenge is to make this structural argument quantitative.

7.2 Required theoretical work

In order of priority:

1. **Fine structure constant structural derivation.** Move from the 0.98 ppm numerical candidate to a role-axiomatic derivation that produces the expression from D-C balance geometry without fitting.
2. **Koide formula structural derivation.** Produce a derivation showing that the quadratic mean form of the Koide formula follows from the tholonic role weights, and identify which physical systems are expected to satisfy it.
3. **Proton/electron mass ratio derivation.** Show why $N_{\text{weight}} \cdot \pi^{D_{\text{axis}}}$ gives the ratio of D-role to C-role masses in the hydrogen tholon.
4. **Three spatial dimensions formal proof.** Prove formally that stable recursive N-D-C structure with volumetric closure requires exactly three spatial dimensions.
5. **Quark mass Koide deviation.** Determine whether a unified renormalization scale or a color-charge correction to the tholonic mapping recovers $Q \approx 2/3$ for quark triplets, or formally restrict the Koide correspondence to the lepton sector.
6. **CKM mixing angle address.** Identify whether the quark mixing hierarchy (Cabibbo angle, b-quark suppression) has a tholonic structural reading.

7.3 Scope of the claim

This paper claims that the tholonic model is a *candidate* alternative to the Standard Model, not a replacement. The distinction matters. A candidate alternative is a framework that (a) is internally consistent, (b) addresses at least some of the target framework's open problems, (c) makes falsifiable predictions that the target framework does not make, and (d) has a specified path toward the open derivations that would, if completed, constitute the strongest available confirmation.

The tholonic model currently satisfies all four conditions. It does not currently satisfy the stronger criterion: it does not yet derive enough Standard Model parameters with sufficient precision to constitute a predictive replacement. That criterion requires completing at least the fine structure constant derivation. Until that is done, the correct characterization is: structurally better in several respects, quantitatively competitive in one, and with a clear agenda for the work required to make the quantitative case conclusive.

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8 Implications

The Standard Model's 19 free parameters and 17 independently postulated fields are not merely aesthetic deficiencies. They represent genuine explanatory gaps: properties of the physical world that the most successful theory in physics treats as given rather than derived. The historical analogy is instructive. Kepler's model of planetary orbits was more predictively accurate than Copernicus's circles; Newton's mechanics explained why Kepler's laws hold. The Standard Model is in Kepler's position: extraordinarily accurate, but explaining rather than deriving the structure it describes.

The tholonic model proposes that the structure underlying the Standard Model is the N-D-C tholonic recursion. The proposal is speculative. But it is internally consistent, makes falsifiable predictions, and addresses several of the Standard Model's structural silences without multiplying free parameters. Its five structural correspondences (three generations, three colors, three spatial dimensions, confinement, Koide 2/3) come at the cost of one additional axiom: the N-D-C triad as the universal unit of stable recursive organization. Whether that additional axiom is worth the explanatory gain is an empirical question, and its answer depends on whether the quantitative derivations in Section 7.2 succeed.

Tesla and Einstein both argued, from different positions, that the field is primary and that what we call particles are stable field configurations. The tholonic model provides a specific structural grammar for that claim: the A&I field is the single underlying field, the tholonic recursion is the organizing principle, and the Standard Model's diverse particle spectrum is the thologram's stable excitation structure. The unified field that Einstein pursued for thirty years is proposed to be this structure. Whether it is that structure is the question the derivations in Section 7.2 will decide.

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9 References

- **Tholonia: The Mechanics of Existential Awareness** by Duncan Stroud, especially Chapters 11 through 15.
- **Paper 1 in this series:** *Emergence of Classical Constants from a Minimal Recursive Triadic Framework.* Derivation of $\pi/4$, ϕ, e, $\sqrt{2}$, $\ln 2$ from tholonic recursion.
- **Paper 3 in this series:** *Minimal Recursive Triadic Framework.* Formal proof that three functional roles are the minimum for non-trivial convergent recursion.
- **Paper 6 in this series:** *The Qualitative Nature of One, Two, and Three: Structural Role Assignment in Minimal Recursive Systems.* Formal basis for the N-D-C integer role assignment.
- **Paper 7 in this series:** *Engineering Toward N: Cambridge Semantics as Empirical Validation of the Tholonic N-D-C Framework.* Independent engineering convergence on the N-D-C structure.
- **Paper 8 in this series:** *The Atom as a Measurable Tholon: From Einstein's Incomplete Electron to a Field-First Ontology.* Atomic correspondences and Standard Model comparison that this paper extends.

- **Koide, Y.** (1982). "A fermion-boson composite model of quarks and leptons." *Physics Letters B* 120, 161–165.
- **Cirel'son, B.S.** (1980). "Quantum generalizations of Bell's inequality." *Letters in Mathematical Physics* 4, 93–100.
- **Feynman, R.P.** (1985). *QED: The Strange Theory of Light and Matter.* Princeton University Press. (Source of Feynman's description of α as "one of the greatest damn mysteries of physics.")
- **Einstein, A.** (1949). "Autobiographical Notes." In *Albert Einstein: Philosopher-Scientist*, ed. P.A. Schilpp. Open Court.
- **Particle Data Group.** (2024). *Review of Particle Physics.* Progress of Theoretical and Experimental Physics 2024, 083C01.
- **CODATA.** (2022). *CODATA Internationally Recommended 2022 Values of the Fundamental Physical Constants.* NIST.
- **Weinberg, S.** (1995). *The Quantum Theory of Fields*, Vol. 1. Cambridge University Press.
- **Gross, D.J., Politzer, H.D., Wilczek, F.** (1973). "Ultraviolet Behavior of Non-Abelian Gauge Theories." *Physical Review Letters* 30, 1343–1346. (Asymptotic freedom, QCD basis for confinement.)
- **ALEPH, DELPHI, L3, OPAL Collaborations.** (2006). "Precision electroweak measurements on the Z resonance." *Physics Reports* 427, 257–454. (LEP neutrino species constraint $N_\nu = 2.9840 \pm 0.0082$.)